

ATC Communications

Radio Techniques 1

Adhering to the following principles in radio communications will help eliminate many of the common errors in airfield driving:

- Don't make the mistake of hearing what you expect or want to hear, rather than your actual ATC instructions
- Ask the Control Tower for clarification when in doubt about an instruction
- Ask the Control Tower to repeat their instruction if you are not sure about what you heard by saying "say again".
- Read back (including runway identifier) all instructions.



Radio Techniques 2

Additionally, the following common radio techniques will make radio communications quicker and error free:

- Prepare first: Your transmission should be well thought out.
- Before keying your transmitter, know what you want to say and check to make sure you are on the proper radio frequency
- Communication with Control Tower should be concise and to the point. For unusual situations or lengthy communications, initial contact should be made prior.
- Acknowledge all instructions with your call sign.
- If any portion of the transmission is "garbled" or "stepped on" do not assume the unheard portion is irrelevant. Request that the transmission be repeated by saying "say again."
- If your call sign is not included in the transmission, don't assume the transmission is for you

- Listen; don't assume the transmission is what you requested.



Radio Techniques 3

Effective tower communications are critical in avoiding a Runway Incursion incident.

Proper phraseology is an essential element of effective tower communications. Specific rules are used to eliminate misunderstanding between you and the tower. Some of those rules are listed below:

- DO NOT use "10" codes, such as "10-4" to acknowledge understanding.
- DO NOT use the words "clear" or "cleared". The term is only used for communication between aircraft and ATC.
Example: "Clear to land" or "Cleared for Take-off".
- ALWAYS use phonetic alphabet to pronounce taxiway location.

Radio Techniques 4

Air Traffic Controllers are busy. When communicating with ATC you should never assume:

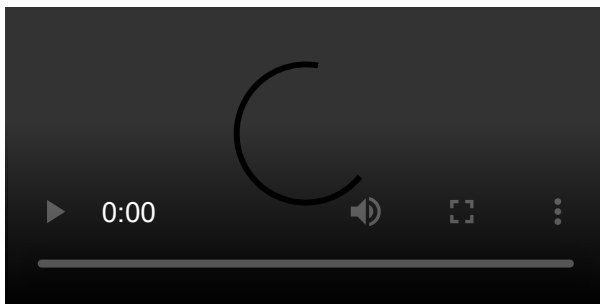
- They know where you are, who you are, or what you are going to do.
- They gave you authorization to cross or enter a runway.
- That you know more than ATC about what aircraft traffic is landing, taking off, or flying in the local area.
- The runway is closed or operations are suspended. Always get ATC approval prior to entering the runway environment.



Radio Techniques 5

Tower communications rules follow a specific format and must be rigidly adhered to. This is not to limit your communication capabilities, but to ensure that both parties understand what is being communicated. The established communication format is as follows:

1. Identify who you are calling.
2. Identify yourself and your location.
3. Communicate specific information or request using proper phraseology.
4. Acknowledge understanding of tower controllers' instructions, by repeating back verbatim the towers instructions.



Video Transcript:

[vehicle] Lackland tower, maintenance 1 9.

[tower] Maintenance 1 9, Lackland tower.

[vehicle] Lackland tower, maintenance 1 9 at the intersection of taxiway bravo and runway 3 6. Requesting permission onto runway 3 6.

[tower] Maintenance 1 9, Lackland tower. Proceed on runway 3 6 and report when off.

[vehicle] Lackland tower, maintenance 1 9. Proceeding onto runway 3 6 from taxiway bravo. Will report when off.

Radio Techniques 6

Once you are approved to enter a controlled movement area you are considered a potential mobile obstacle by the tower.

This is why it is imperative that direct and continuous radio communications be maintained with the tower to avoid becoming a hazard to aircraft movement.

Scan the full length of the runway and the final approaches before entering any runway, even if you have received Control Tower approval to enter the runway.

Note: Vehicles operating on the runway must use rotating beacon lights or hazard warning flashers.



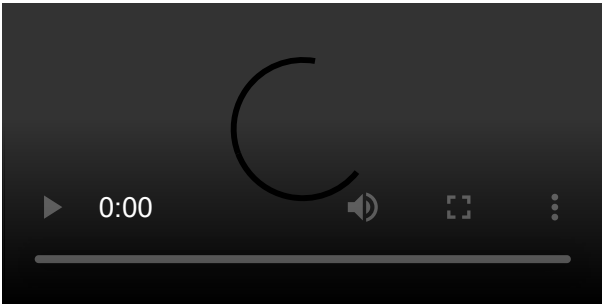
Radio Techniques 7

All vehicles without a radio will be escorted by a vehicle with a radio in direct communication with the Control Tower.

Note: You may not obtain approval onto or across a runway from any source other than the tower. If you are under escort by another vehicle with a radio, pull onto the runway only at the direction of the escorting vehicle.

The escorted vehicle driver must remain with their escort at all times while operating on the runway in the controlled movement area.

The same principle of direct and continuous communication with the tower applies in instances when a vehicle without the means to communicate with the tower and/or by a vehicle operator without a valid AF Form 483.



Video Transcript:

[vehicle] Langley tower, airfield 1.

[tower] Airfield 1, Langley tower.

[vehicle] Langley tower, airfield 1 + 1 at the intersection of taxiway bravo and runway 3 6. Request permission to cross runway 3 6.

[tower] Airfield 1 + 1 proceed onto runway 3 6 from taxiway bravo. And report when off.

[vehicle] Langley tower, airfield 1 + 1 proceeding across runway 3 6 from taxiway bravo. Will report when off.

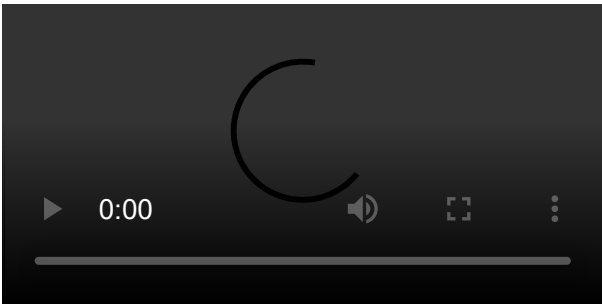
[vehicle] Langley tower, airfield 1 + 1 are off runway 3 6 at taxiway alpha.

Lightgun Techniques 1

Now that you have completed Radio Techniques let's review tower light signals for greater clarity.

Note: During daytime operations glare may reduce your ability to see tower light gun signals.

COLOR	ON THE GROUND
	Cleared to cross, proceed or go
	Stop
	Clear runway or taxiway
	Return to starting point
	Exercise extreme caution

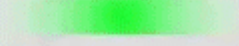


Lightgun Techniques 2

It is important that you follow the ATC light gun instructions exactly as specified on the DAFVA 11-240 placard/decal to prevent a runway incursion incident.

Note: Your vehicle will have an DAFVA 11-240 placard/decal installed with the light gun codes illustrated which indicates actions to be taken.



STEADY GREEN	
CLEARED TO CROSS: PROCEED OR GO	
FLASHING GREEN	
NOT APPLICABLE	
STEADY RED	
STOP	
FLASHING RED	
CLEAR THE TAXIWAY OR RUNWAY	
FLASHING WHITE	
RETURN TO STARTING POINT ON AIRPORT	
ALTERNATING RED/GREEN	
GENERAL WARNING EXERCISE EXTREME CAUTION	

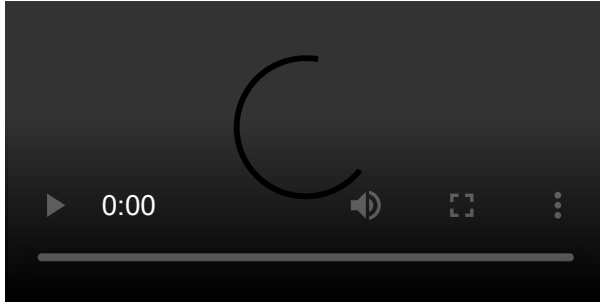
Lightgun Techniques 3

In the unlikely event of radio failure after you have entered the runway, they will attempt to get your attention by flashing the runway edge lights.

If the edge lights begin flashing while you are on the runway, immediately contact the tower and respond to lightgun signals.

Note: Requirements for loss of communication on runways may vary so be familiar with local procedures.

Exit the runway immediately if you're unable to regain radio communication with the control tower.



Summary

In this module you learned effective radio communication techniques, communication procedures for escorted/unescorted runway vehicle entry, and light gun operations during radio failure.